

PROPOSAL OF EXAMINATION PROJECT

AI Models for Physics 2025/2026

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Title: Physics-Informed ML and Domain Adaptation for Gravitational Lensing

Learning method/algorithm used:

Physics-Informed Neural Networks (LensPINN), Vision Transformers (HEALSwIn), Unsupervised Domain Adaptation (ADDA).

Objective: (max 3 lines):

To build a physics-informed model classifying 4-class dark matter substructures in gravitational lenses, adapting from synthetic dataset simulations to real telescope imagery (HSC) using unsupervised domain adaptation.

Based on a previously existing project:

☐ No, it is totally new not based on code of others

☒ Yes, I have partly used code from the following repositories:

url: <https://github.com/ML4SCI/DeepLense>

Reference used:

Link(s):

http://ml4physicalsciences.github.io/2024/files/NeurIPS_ML4PS_2024_78.pdf (LensPINN)

http://ml4physicalsciences.github.io/2025/files/NeurIPS_ML4PS_2025_252.pdf (HEAL-PINN)

<http://arxiv.org/abs/2112.12121> (Domain Adaptation)

<http://arxiv.org/abs/1702.05464> (ADDA)

Needed dataset:

☐ Not needed

☒ Yes: I have original data from a research project (DeepLense synthetic datasets provided by the ML4SCI collaboration)

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() Yes: Published here url: <http://....>